

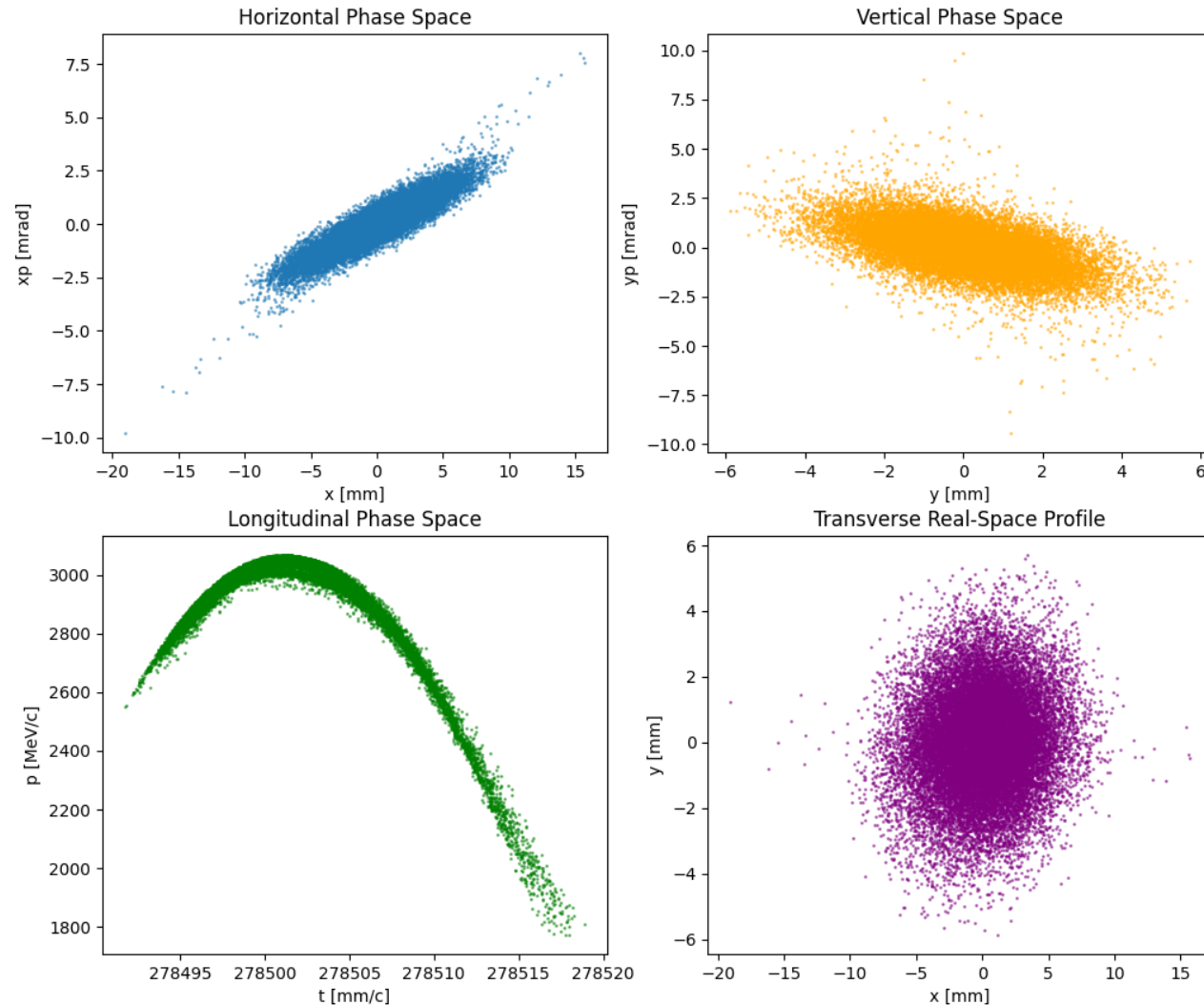
Macroparticle Distribution Simulations

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Overview

- Used the latest macroparticle beam distribution as the injection source
- Analysed the beam phase space and calculated the corresponding emittances
- Normalised the distribution and injected it into the current polarizer ring lattice design
- Evaluated injection efficiency and identified the optimum injection energy through an energy scan
- Performed simulation at the selected optimum energy

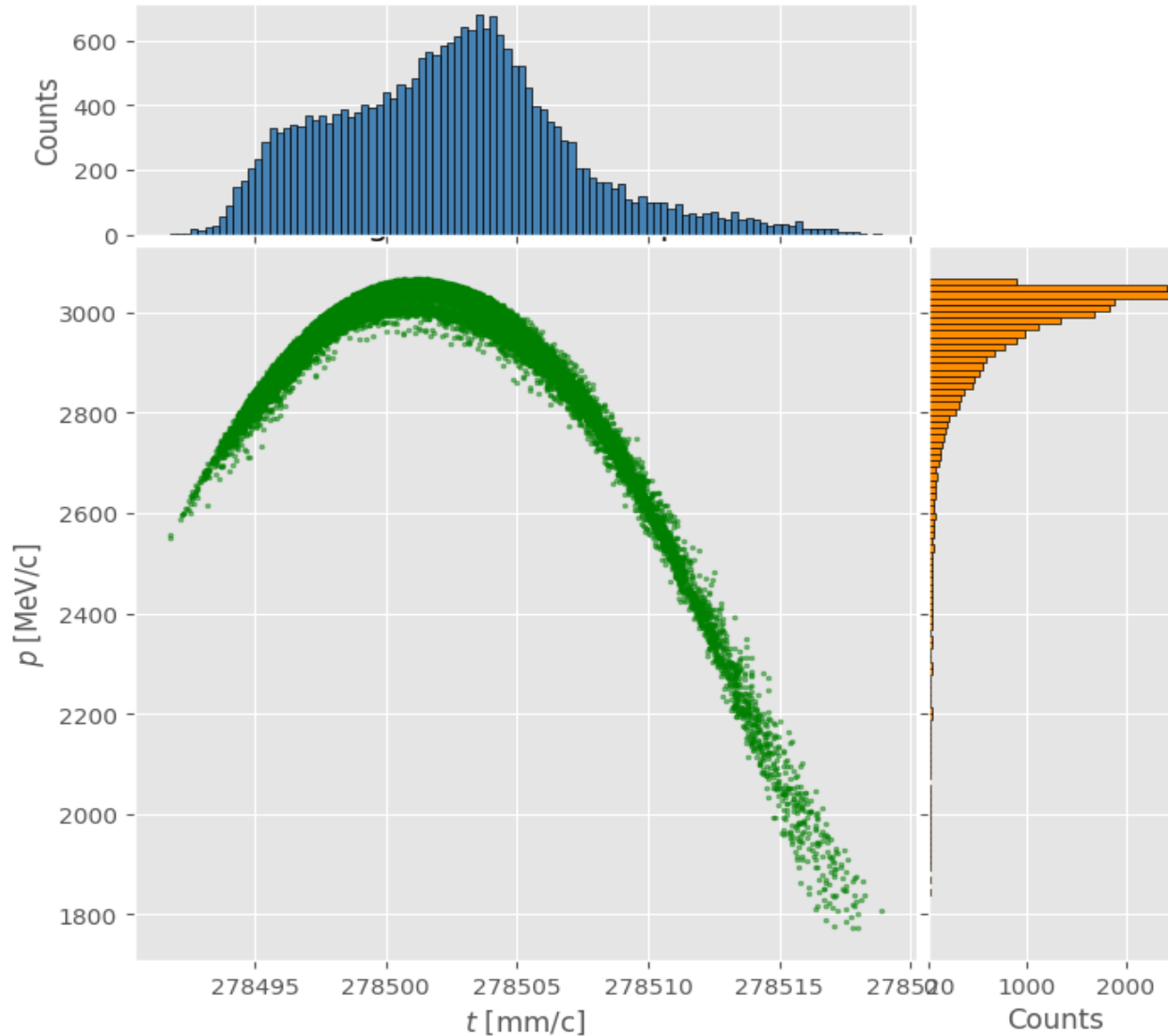
Phase space and emittance



	FCC-ee parameters	Calculated values
Horizontal emittance	2.32 μ m	1.35 μ m
Vertical emittance	2.14 μ m	1.42 μ m

Calculated RMS momentum spread
was 6.48%

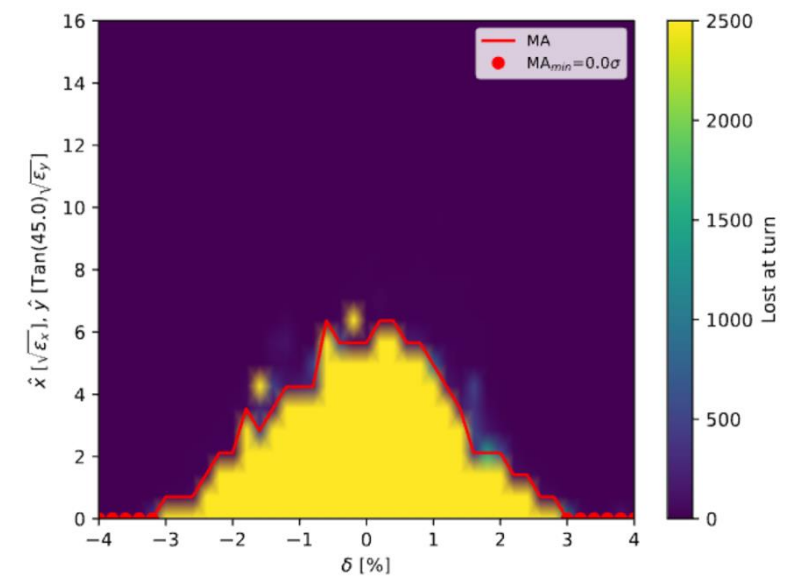
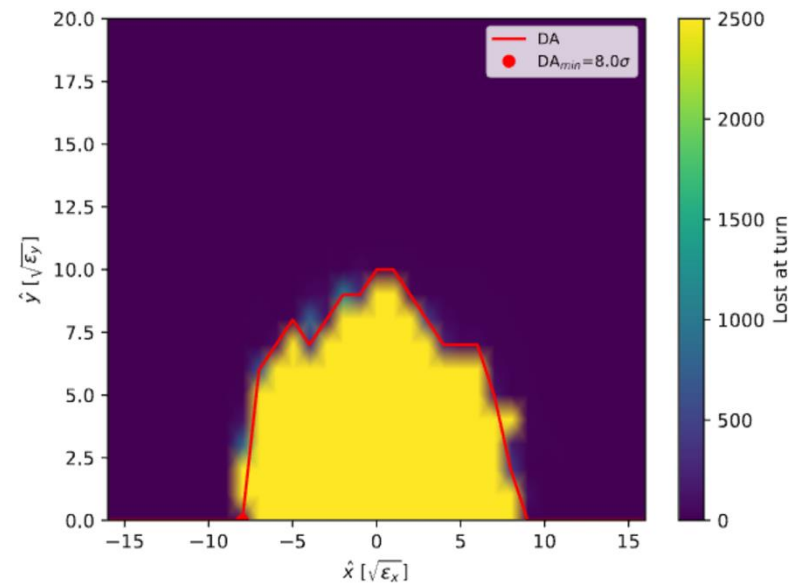
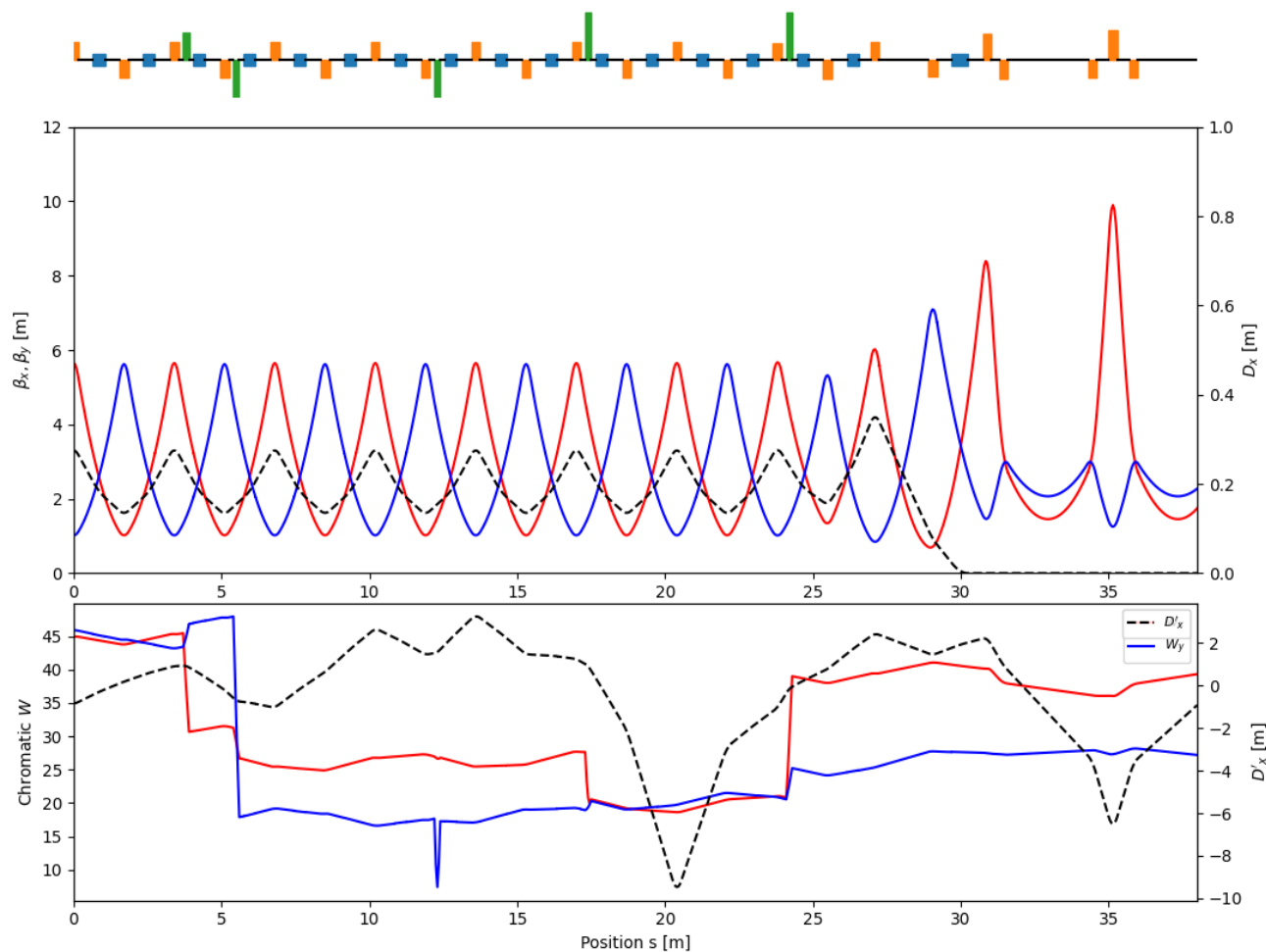
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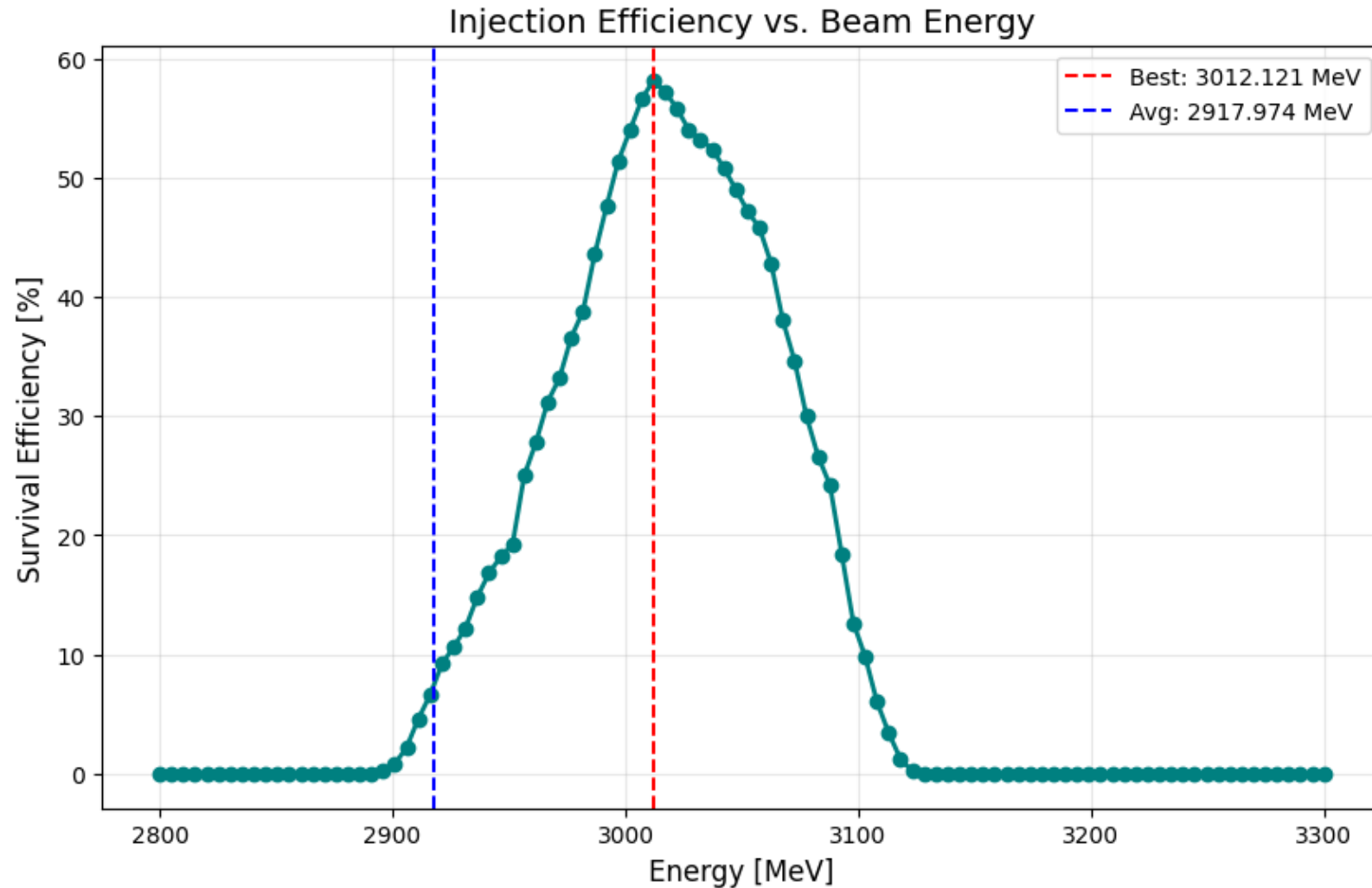
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The polarizer ring



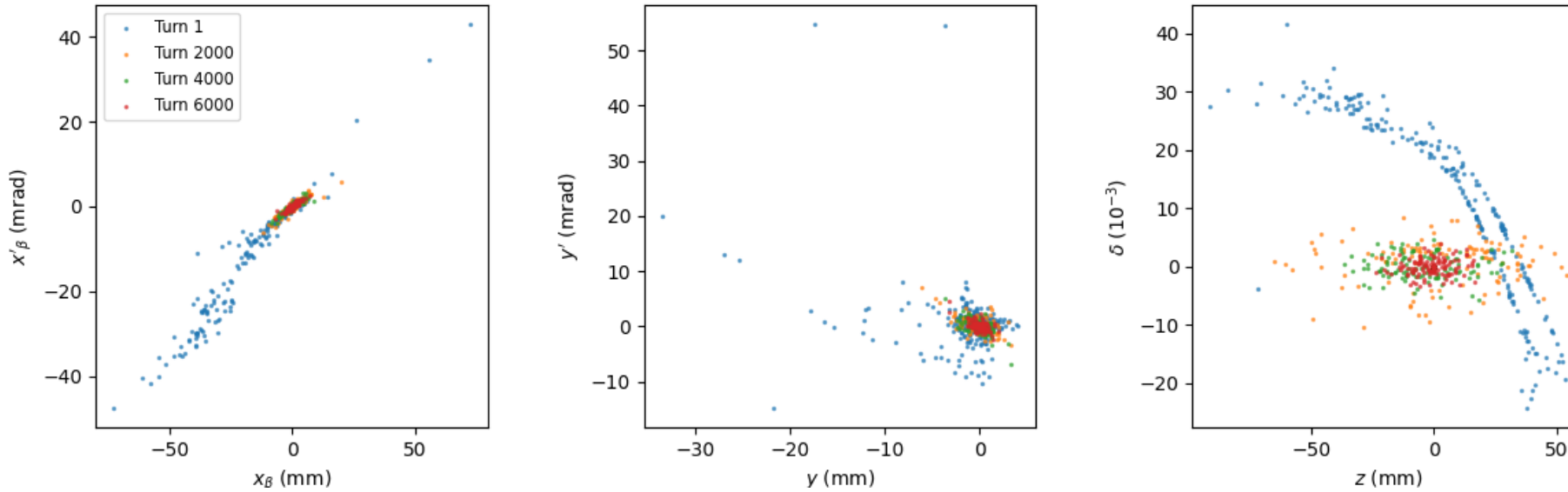
Using momentum spread of 2%

Energy scan for the best efficiency



Initial injection simulation

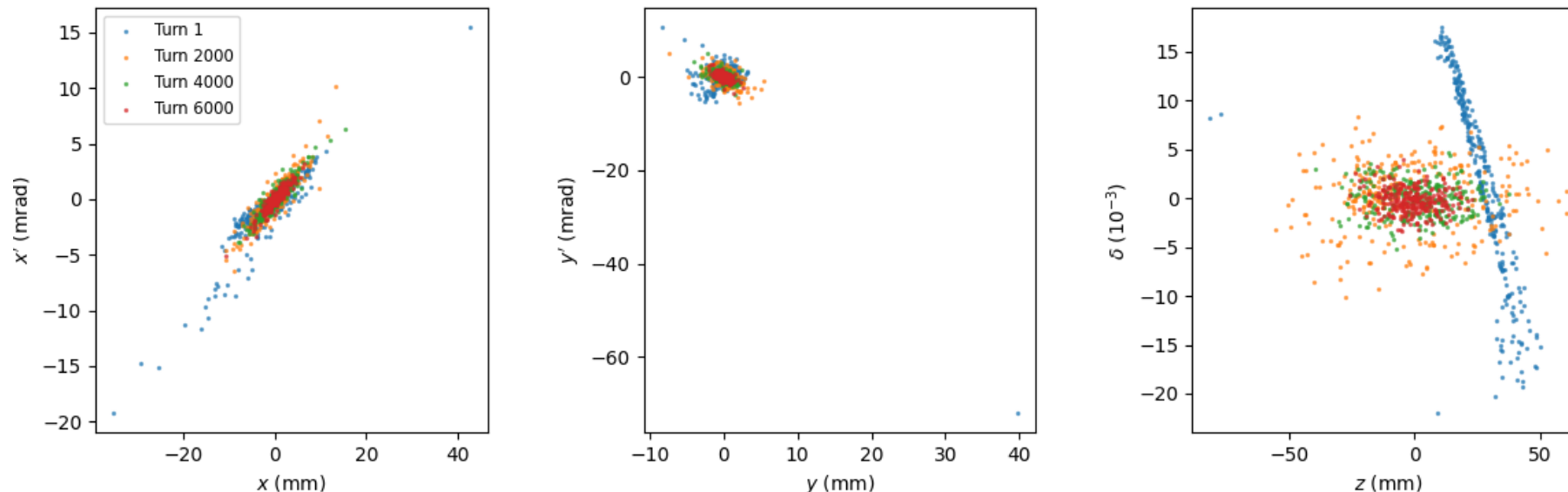
Particle Survival: 117 / 500



- Simulation using the average energy from the distribution at 2.92GeV (this was higher than 2.86GeV)
- ~ 23% of particles survived

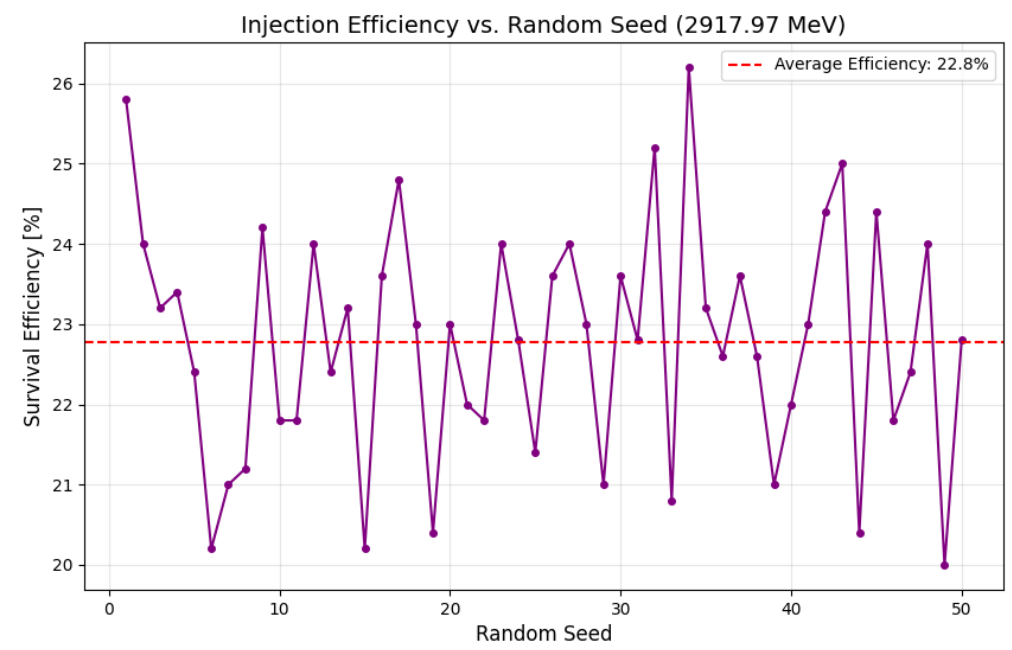
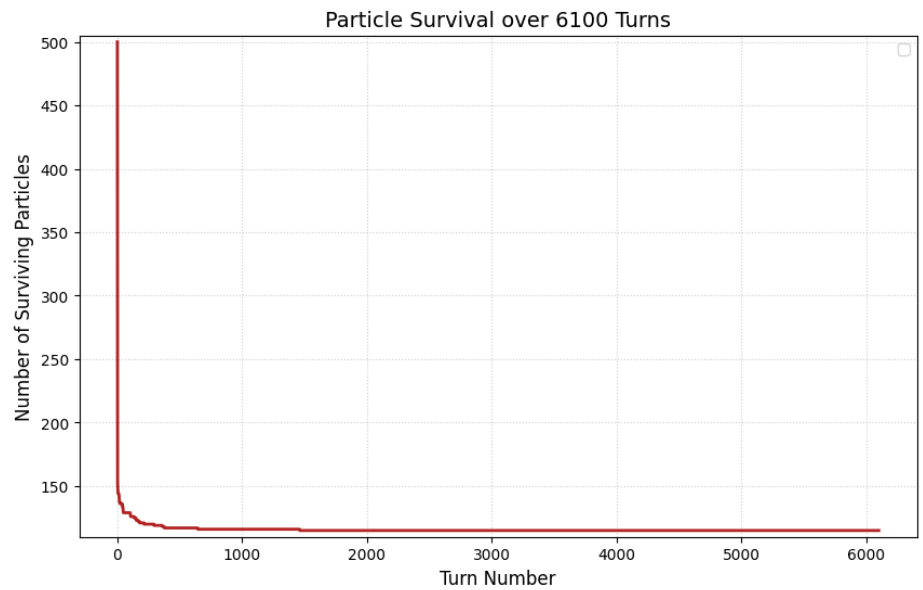
Best injection simulation

Particle Survival: 277 / 500



- Simulation using 3012MeV
- ~55% of particles survived

Initial injection simulation



Best injection simulation

